Why wood first

The talk keeps returning to timber, and that is a choice. Of everything a Vancouver house holds, its wood frame is where reuse is possible and not happening: the most material lost, the most value, and typically the same way of building as the multiplexes replacing it. The hub takes everything; its tests, and my joints, start with wood.

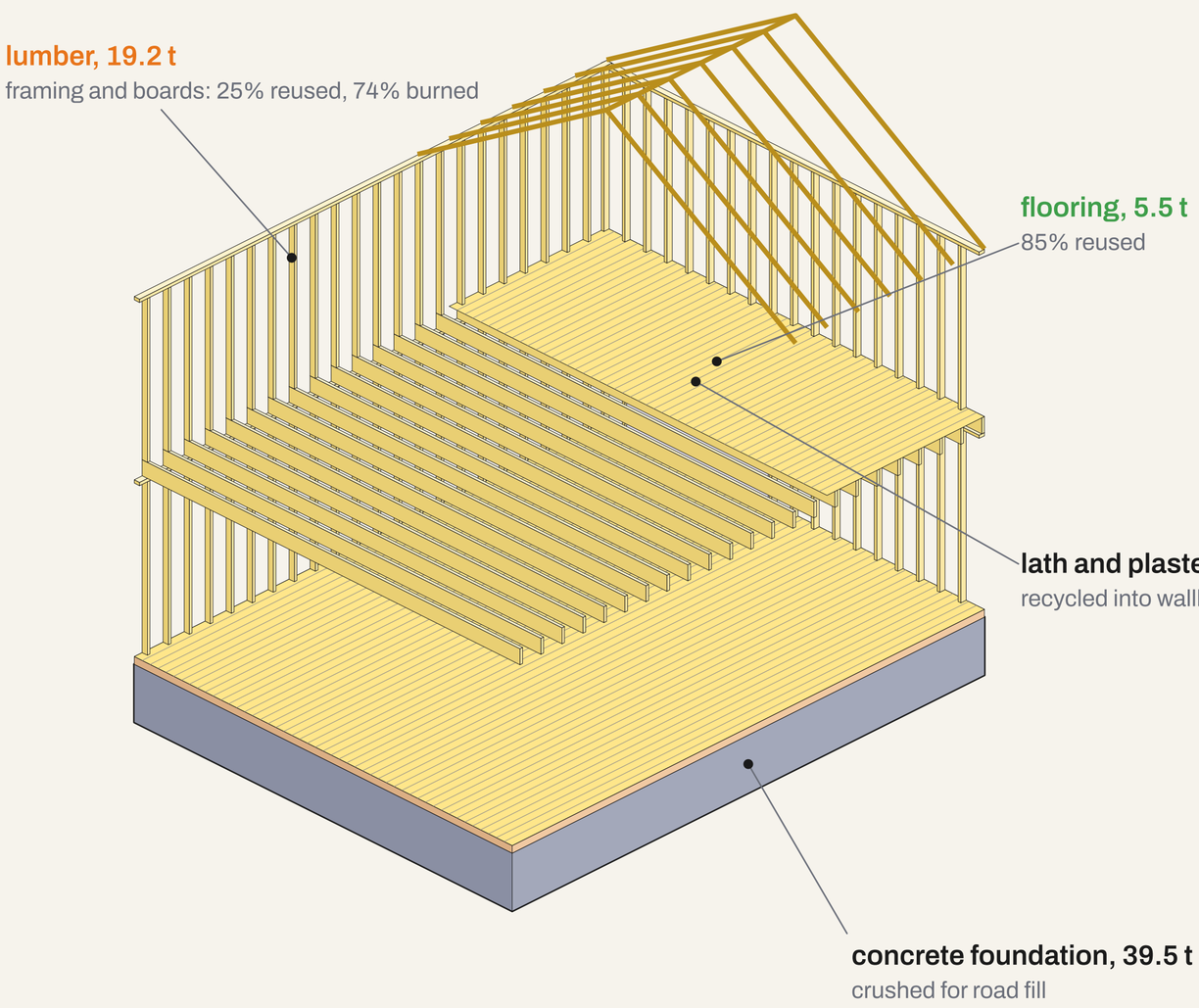
74% of one house's lumber was burned for energy

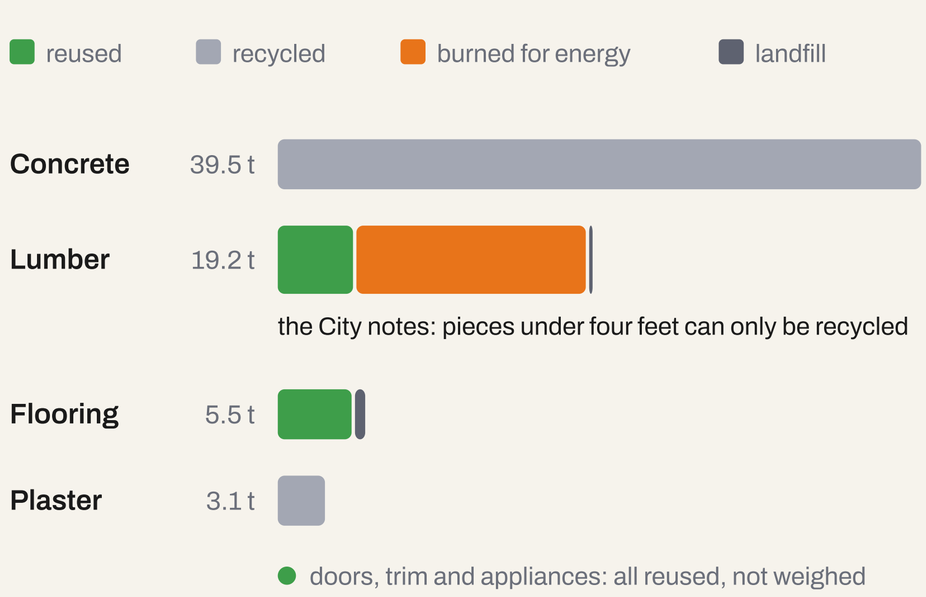
48% of the building waste the region throws away is wood

100,000 t of salvageable lumber in pre-1940 houses, estimated

One house, taken apart by hand

A 1930 house in Dunbar, 17 days in 2019: what each material weighed, and where it went



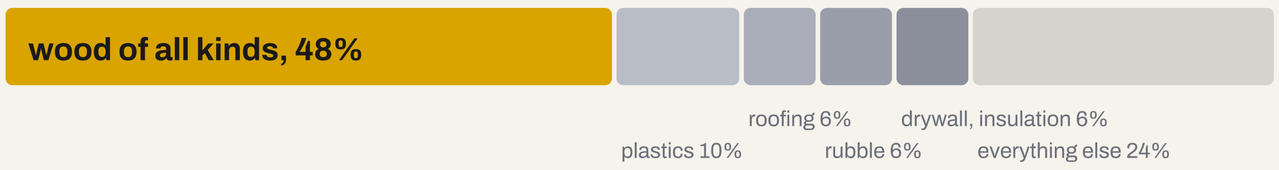


Overall, 98% of the house was kept from landfill, but only 21% was reused. The rest counted as diverted, which is how the by-law measures it. Deconstruction cost $34,990, less $6,000 from selling lumber, against $21,000 to $27,000 to demolish.

The lumber is the gap: heavier than everything but the foundation, likely old growth, and mostly burned. Short lengths are one reason the City names; nails, splits and buyers matter too.

What the region still throws away

Construction and demolition waste disposed of, Metro Vancouver, 2021: about 372,000 t, by weight, from a visual audit



Within the wood, at the Vancouver Landfill



dimensional lumber, 49% of the wood
four fifths of it painted, stained or treated

plywood 21%, particleboard 11%, fibreboard 10%

Concrete is only 2% of what is thrown away, metals 4%: they already have paths, as aggregate and scrap. Wood is what escapes them. Residential demolition was the largest source the study sampled.

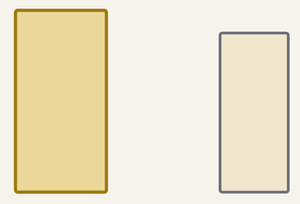
Old growth in old houses

Vancouver's pre-1940 houses, estimated in 2015

135,000 t of dimensional lumber

75% of it salvageable

1,000–1,500 t a year as they come down: framing for 100 to 130 new houses



An old 2×4 is a full two by four inches, denser and drier than new stock. It will not drop into today's framing without design: one more reason to design the joints, not just reuse the sticks.

The same study's caution: old-house framing is about 1% of the City's wood waste. The case for wood first is value and fit, not tonnage.

Why not another material first

material	in the Dunbar house	its path today	why it does not lead
Concrete foundation	39.5 t, the heaviest	crushed for road fill	Cast in place, so it cannot come out as a part; it already has a path.
Lumber	**19.2 t**	**25% reused, 74% burned**	**Leads: the most lost, likely old growth, and multiplexes are typically framed the same way.**
Flooring	5.5 t	85% reused	Already finds buyers when floated or nailed.
Lath and plaster	3.1 t	recycled into wallboard	No reuse path; recycling works.
Doors, trim, appliances	not weighed	resold through ReStores	A market exists.
Roofing, plastics, insulation	not weighed	landfill, some recycling	A recycling problem, not a reuse one.

What wood first leaves out: the carbon of new concrete foundations, which is a question of how the next buildings stand, not of salvage; and windows, whose reuse turns on energy performance. The hub accepts both; the research starts with wood.

What stands in the way

Grading
Here an engineer must sign off each structural reuse. Oregon has let sound salvaged lumber be designed as Douglas fir-larch No. 2 without a stamp since 2017, and Washington's code allows used sawn lumber.

Nails and short lengths
De-nailing is slow, and pieces under four feet are recycled by default.

Paint and treatment
Most landfilled lumber is painted, stained or treated; old paint must be tested for lead.

Sorting and storage
No place to sort and store wood at scale, as AnnaLisa Meyboom and Kaia Nielsen-Roine note: the hub's job.

Piece by piece: the hub's path for wood, gentlest first

Each piece is tested, then takes the first step it can; today, in Dunbar, a quarter took the first step and three quarters the last

1 Reuse whole
tested and graded, back into a frame

2 Re-mill
resawn into flooring, siding and trim

3 Re-join
short pieces joined dry, dowelled, pegged or keyed, into members that come apart: my GP II focus

4 Remanufacture
into strand board, cross-laminated tiles, wood fibre insulation

5 Energy
last, when nothing else works

Sources: City of Vancouver, Deconstruction case study: Dunbar (2019); Metro Vancouver, 2022 C&D Waste Composition Study (Dillon, 2023), a visual audit converted to weight; Teshnizi, UBC SEEDS, 2015; Meyboom and Nielsen-Roine, Canadian Architect, April 2025; City of Portland, June 2026; King County. One house is a case, not an average. Drawing: R. Liu.